

1 Gwanak-ro
Gwanak-gu
Seoul, S. Korea

<https://hyunho111.github.io/>
hun1021405@snu.ac.kr
+82 10-3745-8393
[Google Scholar](#) · [GitHub](#) · [LinkedIn](#)

Hyunho Song

EDUCATION

Seoul National University (SNU)

Mar. 2025 – Feb. 2027 (expected)

M.S. in Future Automotive Mobility

Advisor: Prof. Ayoung Kim

Cumulative GPA: 4.12/4.3

Related coursework: Topics in System Software
Autonomous Driving: Theory and Practices

Seoul National University (SNU)

Mar. 2017 – Feb. 2025

B.S. in Mechanical Engineering

Cumulative GPA: 3.89/4.3

Thesis topic: Effects of dynamic-object filtering on FMCW LiDAR SLAM

Advisor: Prof. Ayoung Kim

Related coursework: Introduction to Robotics
Machine Learning for Mechanical Engineering
Sensor-Based Spatial Intelligence

RESEARCH EXPERIENCE

RPM Robotics Lab, Seoul National University

Mar. 2025 – Present

M.S. Student

- Contributed image-reliability weighting and evaluation to RGB-thermal inertial odometry.
- Built the CNS LiDAR dataset in CARLA for long-term place-recognition evaluation.
- Aligned and aggregated multi-session vector maps for a Hyundai Motor research project.

RPM Robotics Lab, Seoul National University

Aug. 2024 – Feb. 2025

Research Intern

- Estimated and validated extrinsic parameters of the HeRCULES multi-sensor system.

UGV Mapping in Construction Environments

RPM Robotics Lab, Seoul National University

- Integrated obstacle avoidance into a UGV mapping system.

EXTERNAL ACTIVITIES

29th Roboracer Autonomous Racing Competition

Aug. 2026

Team Member, IFAC 2026

- Refined particle-filter localization for high-speed autonomous racing.
- Reduced obstacle-detection latency by restructuring LiDAR scan processing.

HD Hyundai Robotics

Mar. 2024 – Jun. 2024

Intern, Industrial Robot Technology Team

- Performed load analysis and mechanical design for industrial robots.
- Maintained robot design documentation, component specifications, and technical manuals.

ESSYS, FMTC Consortium

Jan. 2024

Intern, Future Mobility Technology Center field training program

- Refined autonomous-driving software through iterative testing.

Mechatronics Contest, Seoul National University

Sep. 2023 – Dec. 2023

Team Member; competition sponsored by HD Hyundai Infracore

- Co-developed an Arduino-based balancing cupholder with IMU feedback and motor control.

Seoul Robot Academy

Jul. 2023 – Aug. 2023

Student, Robot Vision Engineering Training Course

- Applied computer vision to pick-and-place tasks using a collaborative robot.

Korean Patent Attorney Examination

2021

- Passed the first-stage examination.

PUBLICATIONS

1. Seungsang Yun, Hyunsoo Jang, Tai Hyoung Rhee, Hyunho Song, Hyeonjae Gil, and Ayoung Kim. “Cross-Spectral Stereo Inertial Odometry.” *IEEE Robotics and Automation Letters (RA-L)*, vol. 11, no. 9, pp. 10417–10424, 2026.
2. Hyunho Song, Dongjae Lee, Seunghun Oh, Minwoo Jung, and Ayoung Kim. “The City that Never Settles: Simulation-based LiDAR Dataset for Long-Term Place Recognition Under Extreme Structural Changes.” *4th Workshop on Future of Construction, ICRA*, 2025.
3. Hanjun Kim, Minwoo Jung, Chiyun Noh, Sangwoo Jung, Hyunho Song, Wooseong Yang, Hyesu Jang, and Ayoung Kim. “HeRCULES: Heterogeneous Radar Dataset in Complex Urban Environment for Multi-session Radar SLAM.” *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

AWARDS AND HONORS

2025	Best Research Award, 4th Workshop on Future of Construction, ICRA.
2024	First place among five teams, FMTC field training autonomous-driving competition.
2023	Honorable Mention, Mechatronics Contest, Seoul National University.
2017	Valedictorian, Kyunggi High School.

LANGUAGES AND SKILLS

- Korean (native), English (fluent).
- C++, Python, ROS.
- SolidWorks, NX, AutoCAD.